

## Lab 4: Orthogonality and Projections

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 23rd, 2026

Name: \_\_\_\_\_

username: \_\_\_\_\_

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section. To receive credit for Activity 1, you'll need to show your lab TA that all test cases have passed **and** that you have answered the written questions in Task 4. Instructions on how to do this are in the lab notebook.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

### Recap: Projections

- ([Chapter 3.4](#)) The **orthogonal projection** of the vector  $\vec{u}$  onto the vector  $\vec{v}$  is given by

$$\vec{p} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Above, the scalar  $k^* = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}$  was chosen to minimize  $\|\vec{u} - k\vec{v}\|^2$ .

- The vector  $\vec{p}$  is called the orthogonal projection because the resulting error vector,

$$\vec{e} = \vec{u} - \vec{p} = \vec{u} - k^* \vec{v}$$

is orthogonal to  $\vec{v}$ , meaning  $\vec{e} \cdot \vec{v} = 0$ .

### Activity 1: Presidential Speeches and Cosine Similarity

Complete the tasks in the `lab04.ipynb` notebook.

There are two ways to access the supplemental Jupyter Notebook:

- **Option 1 (preferred):** Set up a Jupyter Notebook environment locally, use `git` to clone our [course repository](#), and open `labs/lab04/lab04.ipynb`. For instructions on how to do this, see the [Environment Setup](#) page of the course website.
- **Option 2:** Click [here](#) to open `lab04.ipynb` on DataHub. Before doing so, read the instructions on the [Environment Setup](#) page on how to use the DataHub.

To receive credit for Activity 1, you'll need to show your lab TA that all test cases have passed **and** that you have answered the written questions in Task 4.

## Activity 2: Sum-Difference Orthogonality

Let  $\vec{u} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 5 \end{bmatrix}$  and  $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ -3 \end{bmatrix}$ .

- a) Show that  $\vec{u} + \vec{v}$  and  $\vec{u} - \vec{v}$  are orthogonal.

- b) Now suppose  $\vec{u}, \vec{v} \in \mathbb{R}^n$  are arbitrary vectors with the same number of components. Is it always true that  $\vec{u} + \vec{v}$  and  $\vec{u} - \vec{v}$  are orthogonal?

- If so, prove why.
- If not, specify conditions under which it's guaranteed that  $\vec{u} + \vec{v}$  and  $\vec{u} - \vec{v}$  are orthogonal.

*Hint: Use the distributive property of the dot product, which states that*

$$(\vec{a} + \vec{b}) \cdot (\vec{c} + \vec{d}) = \vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{d} + \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{d}$$

### Activity 3: Orthogonal Projections

Let  $\vec{c} = \begin{bmatrix} 1 \\ 2 \\ -4 \\ 0 \end{bmatrix}$  and  $\vec{d} = \begin{bmatrix} 3 \\ 2 \\ 0 \\ -1 \end{bmatrix}$ .

- a) Find the orthogonal projection of  $\vec{c}$  onto  $\vec{d}$ . Call this vector  $\vec{q}$ .

- b) Find the error vector,  $\vec{r} = \vec{c} - \vec{q}$ . Which vector is  $\vec{r}$  orthogonal to,  $\vec{c}$  or  $\vec{d}$ ? Draw a rough picture of the relationship between  $\vec{c}$ ,  $\vec{d}$ ,  $\vec{q}$ , and  $\vec{r}$ . You may want to review [Chapter 3.4](#).

#### Activity 4: Orthogonal Decomposition with Orthonormal Vectors

Let

$$\vec{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$\vec{v}_1$  and  $\vec{v}_2$  are called **orthonormal**, because they are:

- unit vectors:  $\|\vec{v}_1\| = 1$  and  $\|\vec{v}_2\| = 1$ .
- orthogonal:  $\vec{v}_1 \cdot \vec{v}_2 = 0$ .

Additionally, let  $\vec{u} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$ .

Our goal in this activity is to write  $\vec{u}$  as a linear combination of  $\vec{v}_1$  and  $\vec{v}_2$ . The fact that  $\vec{v}_1$  and  $\vec{v}_2$  are orthonormal makes this simple.

- a) Find  $\vec{u} \cdot \vec{v}_1$  and  $\vec{u} \cdot \vec{v}_2$ . Your answer should involve  $\sqrt{5}$ ; don't use a calculator.

- b) Evaluate

$$(\vec{u} \cdot \vec{v}_1)\vec{v}_1 + (\vec{u} \cdot \vec{v}_2)\vec{v}_2$$

What do you notice?

- c) Why is  $(\vec{u} \cdot \vec{v}_1)\vec{v}_1 + (\vec{u} \cdot \vec{v}_2)\vec{v}_2 = \vec{u}$ ?

*Hint: Start by drawing a picture of  $\vec{u}$ ,  $\vec{v}_1$ , and  $\vec{v}_2$ , and reviewing the “Orthogonal Decomposition” section of Chapter 3.4 and the very last example discussed in yesterday’s lecture.*



### Activity 5: Orthogonal Decomposition with Non-Unit Vectors

- a) Let  $\vec{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$ ,  $\vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$  and  $\vec{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$ . Write  $\vec{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$  as a linear combination of  $\vec{v}_1$ ,  $\vec{v}_2$ , and  $\vec{v}_3$ , and verify that your answer is correct. Note that  $\vec{v}_1$ ,  $\vec{v}_2$ , and  $\vec{v}_3$  are pairwise orthogonal.

- b) In general, suppose that  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$  are **orthogonal** vectors in  $\mathbb{R}^n$ , meaning that  $\vec{v}_i \cdot \vec{v}_j = 0$  for all  $i \neq j$ . **Given that** it is possible to write  $\vec{u}$  as a linear combination of  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ , show that the coefficients of the linear combination

$$\vec{u} = a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_d \vec{v}_d$$

are given by

$$a_i = \frac{\vec{u} \cdot \vec{v}_i}{\vec{v}_i \cdot \vec{v}_i}$$

*Hint: Start by taking the dot product of both sides of the linear combination equation with  $\vec{v}_1$ . What do you notice?*

- c) In Activity 4, why didn't we divide by  $\vec{v}_1 \cdot \vec{v}_1$  (and  $\vec{v}_2 \cdot \vec{v}_2$ ) when finding the coefficients?

## Activity 6: A Plane from Spanning Vectors

This activity previews ideas that we will explore in upcoming lectures.

An important idea from [Chapter 4.1](#) is that two non-parallel vectors in  $\mathbb{R}^n$  (where  $n \geq 2$ ) span a plane in  $n$ -dimensional space. Here, we'll show you how to find the equation of such a plane, given two vectors in  $\mathbb{R}^3$ . This is also touched on in [Chapter 4.4](#).

Let

$$\vec{u} = \begin{bmatrix} 5 \\ -7 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}, \quad P = \text{span}(\{\vec{u}, \vec{v}\})$$

- a) Find a nonzero vector  $\vec{n}$  that is orthogonal to both  $\vec{u}$  and  $\vec{v}$ . Verify your answer using dot products.

- b) Suppose your vector is  $\vec{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ . Verify that both  $\vec{u}$  and  $\vec{v}$  satisfy

$$ax + by + cz = 0$$

Then, graph your equation on Desmos, [desmos.com/3d](https://desmos.com/3d).